Ground State energy Calculation for LiH using VQE

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Background

In this project, the ground state energy of LiH was calculated using the Variational Quantum Eigensolver (VQE) method. The cost function for this program was the LiH hamiltonian procured from the website qumulator.org. Four different parametrized circuits or ansatz were employed to optimize the ground state energy and their relative closeness to the literature. Three of the ansatz used yielded pretty similar results, which was very close to the literature value for the ground state energy of -7.91957 Eh for LiH found using Reduced Quantum Monte-Carlo Calculations [1].

In this project, the adjustable parameter was a rotation angle, θ. Changing this angle changed the state prepared by the circuit and therefore its expected energy.

Method:

1. The Jordan-Wigner mapped Hamiltonian was procured from qumulator.org. This Hamiltonian served as the trial state or cost function.

2. The trial state was then operated on using three different ansatz. The purpose of the ansatz was to entangle the qubits in order to steer them towards the desired state. The three ansatz were as follows :
    - The first ansatz applied a rotation y operation to the control qubit, followed by a CNOT gate. This yielded a minimum energy value of
    - The second ansatz applied rotation y gates to both of the qubits, followed by a CNOT gate. This yielded a minimum energy value which was very close to the value obtained using the first ansatz.
    - The third ansatz applied rotation y gates to one of the qubits, followed by a CNOT gate. A rotation y gate was then applied to the second qubit, this yielded a minimum energy value which was very close to the value obtained using the first two ansatz.
    - The fourth ansatz applied two rotation x gates followed by two rotation y gates to the two qubits, then a CNOT gate was applied to the q1 qubit. This ansatz yielded a result with the largest deviation from the literature value.

3. The system was then put through the classical optimizing algorithm COBLYA.

4. COBLYA compares the fidelity of our results to the expected values and switches the ansatz until a desired fidelity

5. Plotted the VQE optimized convergence curve for three different ansatz.

Founded results:

1. Optimized angle: -0.297
2. Optimized energy: −7.88110339 Ha
3. Exact energy: -7.91957 Ha

Conclusion:

The VQE simulation found an angle that lowered the expected energy of the supplied two-qubit LiH model. This demonstrates how a classical optimizer can adjust a quantum circuit to search for a low-energy state.

References:

[1]: Ground state property calculations of LiH_n complexes using IBM Qiskit's quantum simulator : https://docs.google.com/document/d/1NjtVi480Qt0n6Pu9NstR_qsQqgLuTjzixI3C3haV-Yc/edit?tab=t.0

[2]: State Preparation on Quantum Computers via Quantum Steering: **https://docs.google.com/document/d/1Dt3ewa8HcPTI_V7Z-tWMR0hDIEYN0Ef-c4QymsHNrRg/edit?tab=t.0**

[3]: Variational Quantum Algorithms via Measurement-Induced Passive Steering: **https://docs.google.com/document/d/1PNp_p3Krl9vP6ZPL7PoBrhvyj1SN185o9JEs5b11ZDM/edit?tab=t.0**

[4]:Variational Quantum Algorithms via Measurement-Induced Passive Steering: https://docs.google.com/document/d/1K-PbWc2hAexXgyiGmFMtAjr77lBvdKa1SrwsvH4lm0k/edit?tab=t.0